

TECHNICAL DRAWING APPLICATIONS

Maximum Marks: 100

Time allowed: Three hours

1. *Answers to this Paper must be written on the paper provided separately.*
 2. *You will **not** be allowed to write during the first 15 minutes.*
 3. *This time is to be spent in reading the question paper.*
 4. ***The time given at the head of this Paper is the time allowed for writing the answers.***
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5. *Attempt **five** questions in all.*
 6. *You must attempt **three** questions from **Section A** and **two** questions from **Section B**.*
 7. *Each section should be answered on a separate paper.*
 8. *All questions must be answered in full scale.*
 9. *All construction lines must be shown.*
 10. *All dimensions are in millimetres unless specified otherwise.*
 11. *The intended marks for questions or parts of questions are given in brackets[].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 7 printed pages and 1 blank page.

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Turn over

SECTION A (48 Marks)

*Answer any **three** questions from this Section.*

Question 1

- (i) Draw a circle of radius 37 mm. Construct a regular pentagon inside this circle such that all the vertices of the pentagon lie on the circumference of the circle. [6]

Use of protractor is not permitted.

- (ii) Construct an ELLIPSE by the **CONCENTRIC CIRCLES METHOD**. [10]

Given: Major Axis = 135 mm

Minor Axis = 80 mm

Question 2

In a working drawing, a line measuring 6 cm was marked as 12 m. Calculate the Representative Fraction (R.F.) of the scale. [16]

Using this R.F. draw a DIAGONAL SCALE long enough to measure up to 30 m.

Using this scale construct a pentagon and hexagon on opposite sides of a common base measuring 14.3 m.

Question 3

Refer to **Figure 1** given below. Copy the given template showing all the construction lines needed to find the various centres. [16]

(Do not insert any dimensions.)

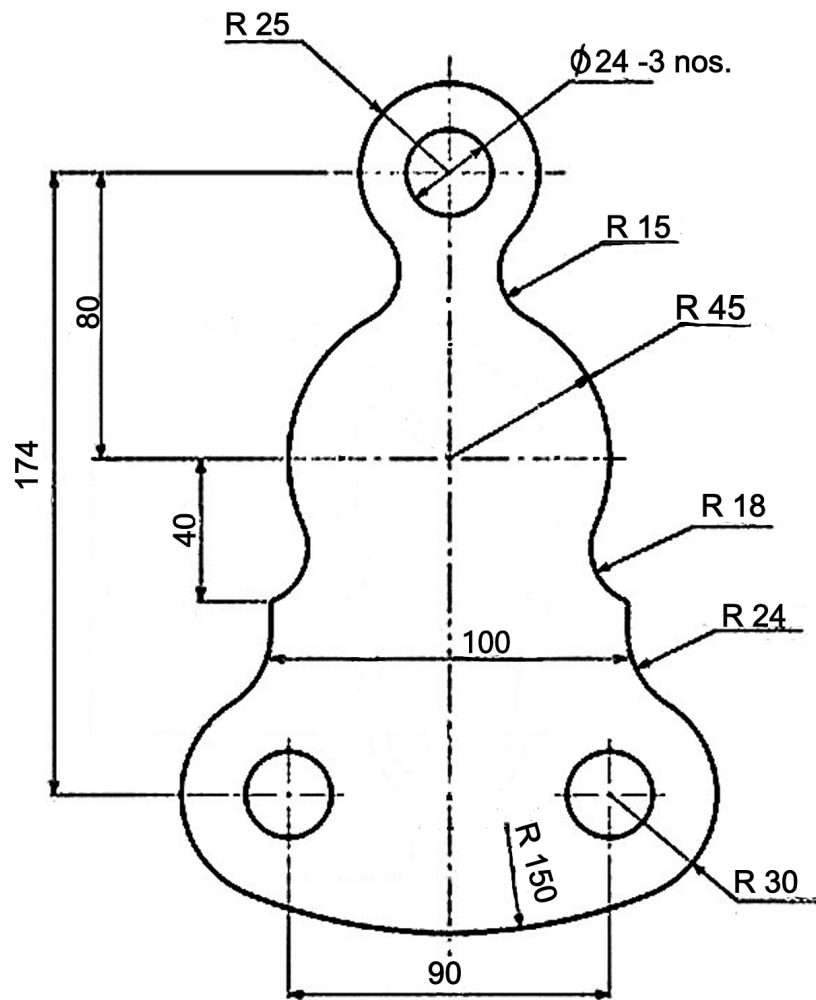


Figure 1

Question 4

Refer to **Figure 2** given below. It shows the Front View (F.V.) and the Right Hand Side View (R.H.S.V.) of an object in the **THIRD ANGLE METHOD OF PROJECTION**. [16]

Draw the **OBLIQUE VIEW** when the receding angle is inclined at 45° to the horizontal.

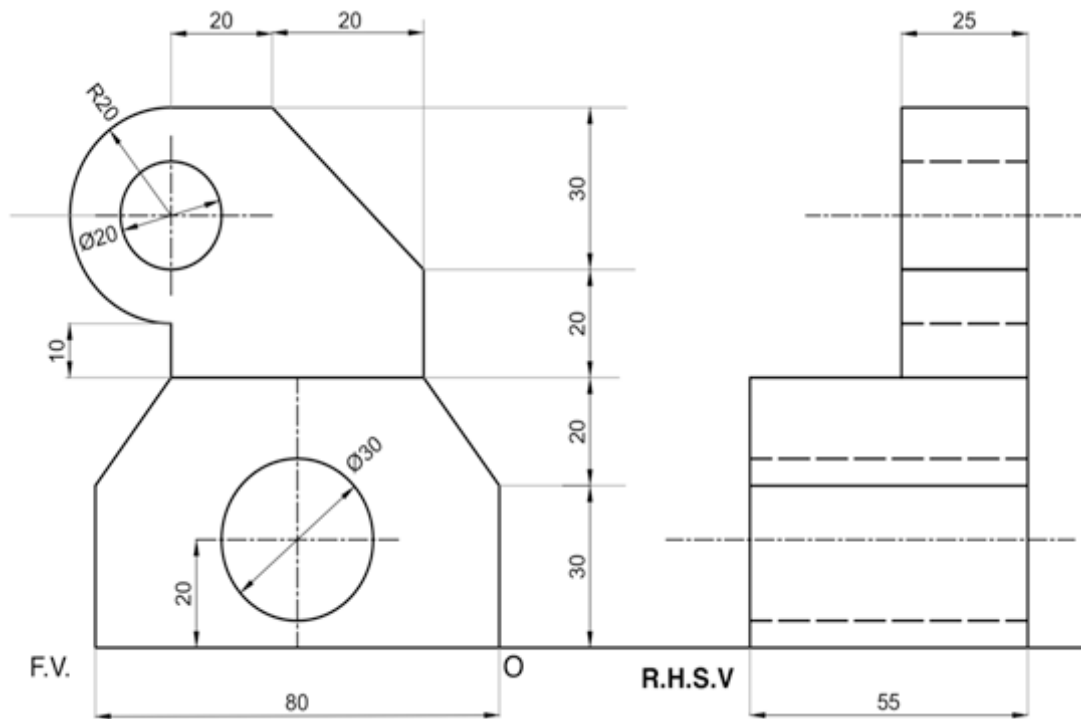


Figure 2

Question 5

Draw Front View (F.V.), Top View (T.V.), Left Hand Side View (L.H.S.V.) and [16]
development of the lateral surface of a pentagonal pyramid whose axis is perpendicular
to the Horizontal Plane (H.P.) and parallel to the Vertical Plane (V.P.). One side of the
base is parallel to the Vertical Plane (V.P.).

Use the **FIRST ANGLE METHOD OF PROJECTION**.

Given side of base = 50 mm and length of axis = 70 mm

SECTION B (52 Marks)

Answer any two questions from this section.

Question 6

Refer to **Figure 3** given below. Copy the given Isometric figure. Show all constructions [26]
neatly. (*Do not insert any dimension.*)

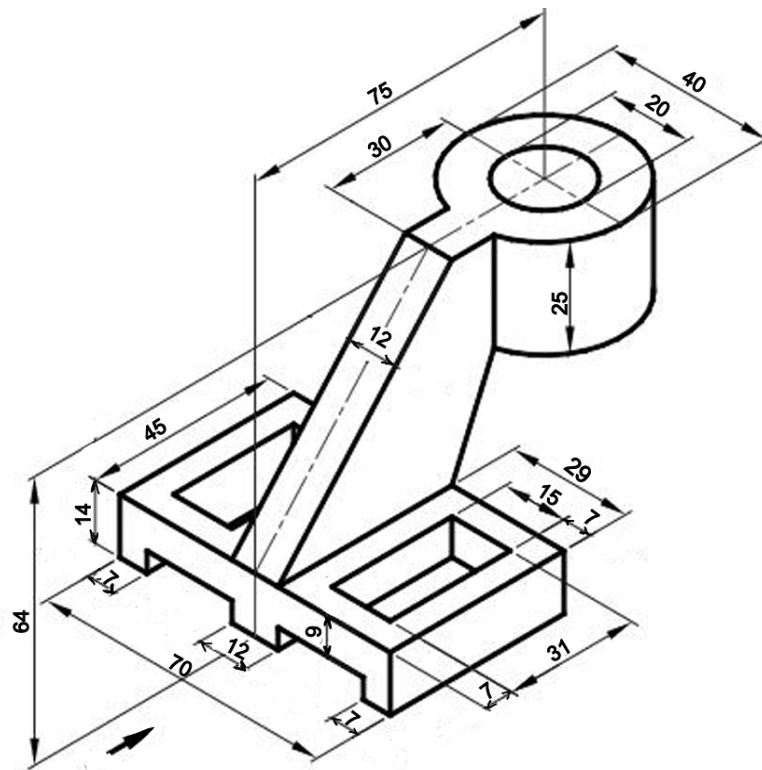


Figure 3

Question 7

Refer to **Figure 4** given below.

[26]

Draw:

- (i) Front View (direction arrow X)
- (ii) Top View
- (iii) Sectional Right-Hand Side View (Along A – A)

Use **FIRST ANGLE METHOD OF PROJECTION.**

(Insert any six dimensions.)

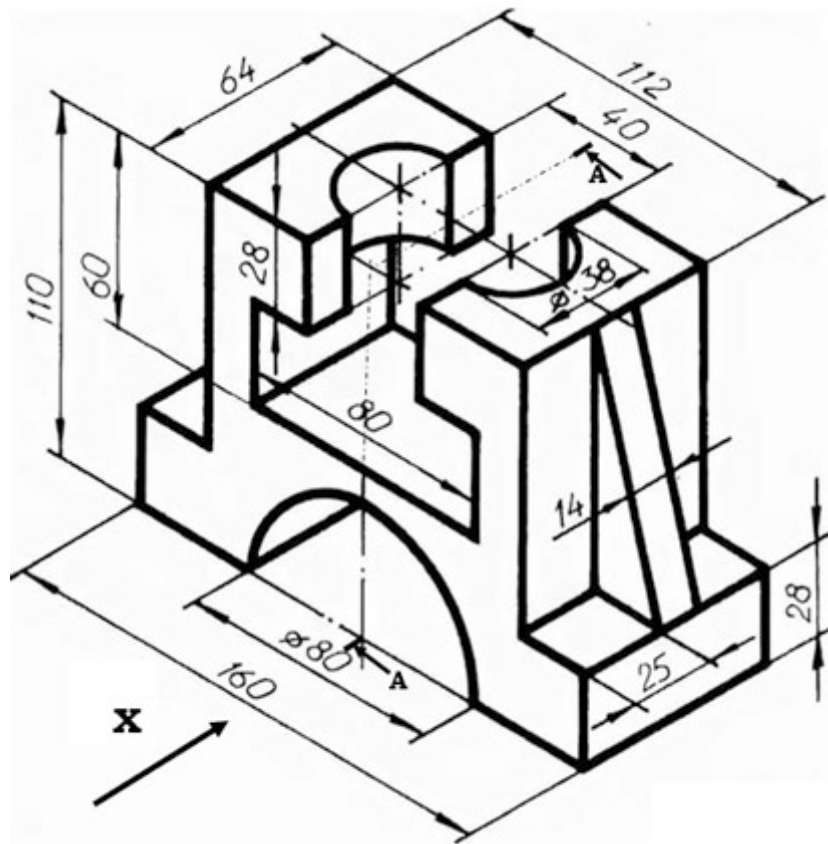


Figure 4

Question 8

Refer to **Figure 5** given below. It shows the Front View (F.V.) and Top View (T.V.) of a [26]
square prism in **FIRST ANGLE METHOD OF PROJECTION**. It is cut by a section
plane as shown in the figure.

Given: Side of base = 30 mm and length of axis = 60 mm

Use **FIRST ANGLE METHOD OF PROJECTION**.

Draw the:

- (i) Front view
- (ii) Sectional Top view
- (iii) Development of lateral surface of part P
- (iv) Auxiliary Top view with true shape of section

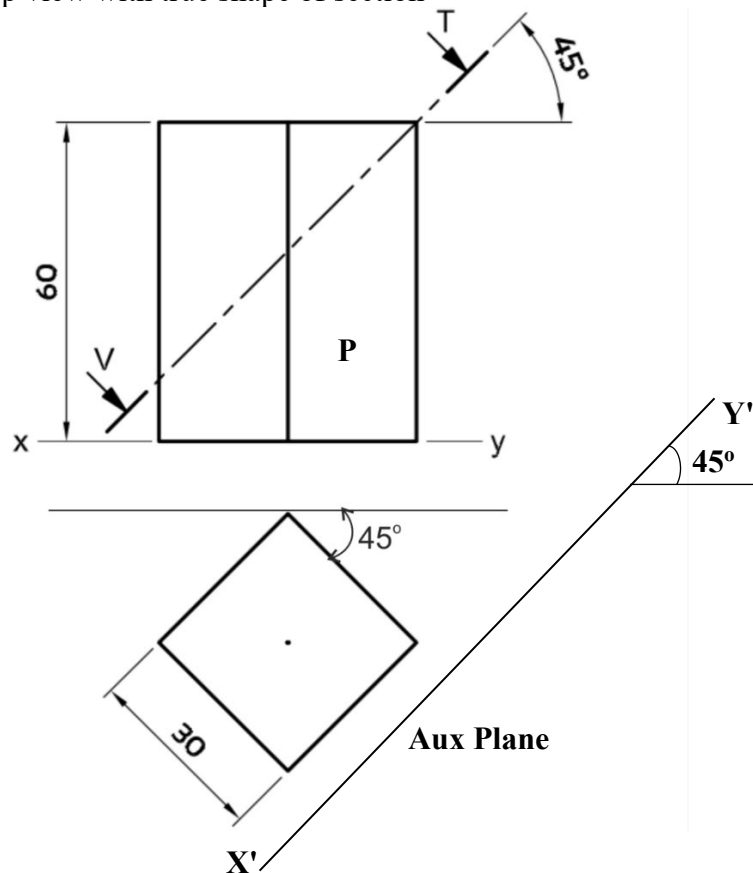


Figure 5